🪰 *Drosophila melanogaster* vs *Bactrocera oleae* — Comparative Summary

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🧬 Phylogeny

- Both belong to **Order Diptera (Suborder Brachycera)**.
- Diverged approximately **70 million years ago (MYA)**.
- They are **more closely related to each other** than either is to ***Aedes aegypti*** (~150 MYA divergence).

🥚 Ovary Maturation

- In both species, ovaries are **immature at eclosion**.
- ***D. melanogaster*** reaches full maturity in **2–3 days** (requires yeast protein).
- ***B. oleae*** matures in **6–10 days** (requires external protein sources).

🍱 Diet

- Both need **sugars for energy** and **proteins for oogenesis**.
- *Drosophila* uses **yeast** as its primary protein source.
- *Bactrocera* relies on **microbial or natural protein sources** (e.g., bird droppings, baits).

⚥ Sexual Dimorphism

- **Females are larger** in both species, with increased abdominal volume for egg production.
- *Drosophila* males have **sex combs**; *Bactrocera* females possess a **strong ovipositor** for fruit penetration.

🐛 Development & Lifespan

Species	Egg → Adult	Adult Lifespan (@25°C)
Drosophila melanogaster	~10 days	30–50 days
Bactrocera oleae	~30–40 days	1–2 months

🧠 Reproduction & Fecundity

- Regulated by **juvenile hormone (JH)**, **insulin/TOR**, and **ecdysone** pathways.
- ***D. melanogaster***: early maturity, **high fecundity (~1000+ eggs/female)**.
- ***B. oleae***: delayed maturity, **moderate fecundity (~300–500 eggs/female)**.
- In both, **oviposition onset depends on post-eclosion protein intake**.

Summary

- Both are evolutionarily close fruit flies sharing **homologous endocrine, developmental, and reproductive mechanisms**.
- ***D. melanogaster***: faster development, higher fecundity, saprophytic lifestyle.
- ***B. oleae***: slower development, host-specific (olive fruits), lower fecundity, field-adapted.